

The design and implementation of a deep reinforcement learning and quantum finance theory-inspired portfolio investment management system

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ABSTRACT

Deep Learning (DL) and Reinforcement Learning (RL) are common machine learning techniques used in automatic trading, notwithstanding, RL is deficient in portfolio investment in terms of funds distribution, potential loss control, profit maximization, and examine undetected environment. This paper proposed an intelligent Quantum Finance-based portfolio investment system (QFPIS), which is a combination of Deep Reinforcement Learning (DRL) with Quantum Finance Theory (QFT) to improve these conditions. There are two major agents embodied in the system: 1) a trading agent based on Deep Deterministic Policy Gradient algorithm to determine investment weighting for different financial products by generating continuous actions; 2) an intelligent agent based on Policy Gradient (PG) algorithm to enact risk control and determine whether to hold current orders by producing discrete actions depend on daily Quantum Price Levels (QPLs). The advantages of incorporating a two-agents system design are to devise stable and realistic fund allocation for different products in portfolio. Experiment results had shown robustness, flexibility, and profitability on a series of forex products and the U.S. stocks in the back-testing phase as compared to other RL trading systems.

1. Introduction

1.1. Research purpose and significance

Time series study is a technique used by researchers under the assumption that are generated by a linear process and consideration of model generation process, but complex, highly noisy, dynamic, non-linear, non-parametric, and even chaotic are typical characteristics of financial time series (Si & Yin, 2013). Deep Learning (DL) is a common machine-learning method used in financial time series forecasting and modeling (Heaton, Polson, & Witte, 2016) for below advantages:

- Pre-process all information related to the expected prediction results as model input.
- Fit nonlinear complex relationships between input variables to improve samples fitting ratio effectively.
- Avoid over-fitting problems when the model consists of simple structures.

Financial analysis has been used successfully in trend movement and

specific prices forecasts regardless of models, input and output types. However, a key issue is how investors can transform the predicted information into profits, that is to design an effective trading system to formulate profitable and low-risk trading strategies by analyzing the predicted information given by forecast system is a challenge to financial engineering (Wang, Zhang, Tang, Wu, & Xiong, 2019). Financial modeling and machine learning methods can be expedited, but the former relied highly on various basic financial assumptions with oversimplification, and limited market prediction accuracy by the latter are inadequate to automatic trading objective by experienced traders and can only be used as a reference.

Reinforcement learning (RL) is a well-developed machine learning method, containing agents to maintain interaction with the environment and to learn an optimal policy by trial-and-error for sequential decision-making problems becomes a possible solution to these shortcomings. (Bao & Liu, 2019).

1.2. Brief description of the model proposed

Financial transactions characteristics are considered when applying

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Deep Reinforcement Learning (DRL) methods:

- Traditional Markov Decision Process model (MDP) is generic, hence researchers are required to devise a detailed definition series representation for RL, to include state (observation) space, action space, and reward function to determine model quality and profitable trading strategies.
- RL agents' individualities are designed for specific game playing and robot commands, hence intelligent trading agents are required to interact with the environment to optimize rewards, manage potential transactional loss with a reasonable risk-control mechanism.

QFPIS is a Deep Reinforcement Learning framework for portfolio management based on the above discussion to formulate the trading strategies directly and maximize the expected accumulative return through weighting adjustments on each financial product contained in portfolio. It embodied two parts:

A trading decision agent based on Deep Deterministic Policy Gradient (DDPG) algorithm basis to generate appropriate weighting vectors.

A risk-control agent based on Policy Gradient (PG) algorithm. The training dataset for PG agent has state space based on DDPG agent, its PG agent function is to select a series of key Quantum Price Levels (QPLs) support or resistance levels to activate or close an order. QPL is a key risk-control mechanism derived from theories of Quantum Finance (QF) theories.

2. Related work

2.1. Theories of Quantum Finance

Reinforcement learning (RL) is a machine learning technique that allows an agent to learn, interact with its environment and receive feedback in the form of rewards or penalties. It has significant attention in investment portfolios management in recent years. Moody and Saffell's (2001) introduced a direct reinforcement learning algorithm called Action-Dependent Reinforcement Learning (ADRL) to learn effective trading strategies in financial markets. It revealed that ADRL algorithm consistently outperformed traditional trading strategies with higher risk-adjusted returns. Deng et al. (2017) extended this work by incorporating deep learning into reinforcement learning framework to proposed Deep Direct Reinforcement Learning (DDRL) model, comprising of deep learning for feature representation and direct reinforcement learning to optimize trading strategies. The results revealed that the DDRL model outperformed traditional trading strategies and other machine learning models in financial markets. Jiang, Xu, & Liang (2017) developed a deep reinforcement learning framework specifically for portfolio management called Financial Portfolio Management using Deep Reinforcement Learning (FPM-DRL) comprising of convolutional neural networks (CNNs) and long short-term memory (LSTM) networks to learn optimal trading strategies from historical price data. This framework achieved superior risk-adjusted returns than other state-of-the-art algorithms. Gao and Huang (2019) introduced a novel deep reinforcement learning approach to portfolio management by incorporating a multi-agent framework. Each asset in the portfolio is as an agent to learn and interact with other agents to optimize portfolio overall performance. It showed effectiveness in various risk-return profiles and improve portfolio diversification.

Lee (Lee, 2020) introduced Quantum Price Level (QPL) based on Quantum Field Theory and Anharmonic Oscillatory Model (AOM) adaptable in financial prediction models. Compared with traditional market features including OPEN, HIGH, LOW, CLOSE, volume and indicators like MACD (Moving Average Convergence/Divergence), QPL is to regard distinctive financial instruments dynamics in different markets in quantum finance theory as a whole as particles in physical area where

wave-particle duality is its major characteristics. Due to Quantum Energy Fields (so-called QPL) operate like equilibrium levels, they affect and restrict intrinsic dynamics and motion of these financial particles. Therefore, trading strategies are devised based on QPL to signify support and resistance lines methods incorporated by traders (Meng et al., 2015). In addition, QPLs calculation underwent a systematic process as follows:

Model financial products' price movements in secondary financial market with Quantum Anharmonic Finance Oscillator (QAHO). Solve Quantum Finance Schrödinger Equation (QFSE) to compute Quantum energy levels for Financial Particles (QFPs) activate in each financial market directly – the Quantum Price Level (Lee, 2020).

2.2. Deep reinforcement learning

Deep Reinforcement Learning has been applied to many regions in finance (Fischer, 2018). Trading is one of the most common and challenging applications. There are two mainstreams by utilizing DRL in trading. The first one is to generate discrete trading decision to apply in trading strategies, represented by (Liu, Xiong, Zhong, Yang, & Walid, 2022), using DDPG to generate buying/holding/selling decisions. An advanced version (Yang, Liu, Zhong, & Walid, 2020) ensembling PPO, A2C and DDPG yielded better performance. (Liu, Liu, Zhao, Pan, & Liu, 2020) use DRL and Imitation Learning to create discrete trading actions as well but limiting action to long/short position by imitating human trading behavior. The second direction is generating portfolio weight directly, which is continuous action. Jiang and his teammates (Jiang et al., 2017) introduced a DRL framework based on DDPG to allocate portfolio automatically. A combination of adversarial training method with PG algorithm (Liang et al., 2018) enabled an effective and efficient portfolio allocation process. Portfolio allocation by DRL is also applied in Cryptocurrency trading and produce satisfactory profit (Jiang & Liang, 2017).

Furthering the application of reinforcement learning in finance by (Liu, Yang, Chen, Zhang, Yang, Xiao, & Wang, 2022), they developed the first open-source financial reinforcement learning framework with three-layer architecture that can support different RL algorithms and trading environment, which can help researchers and developers fast deploy RL applications in financial markets. QFPIS focuses on portfolio weight allocation with the combination of QF-based risk control mechanism, where the risk control is neglected in these previous works.

3. Methodology

3.1. Comparison of quantum anharmonic oscillator model (QAOM) and path integral model

Two mainstream models are used in Quantum Finance: 1) quantum anharmonic oscillator 2) path integral model are compared before deducing QPL. The Hamiltonian is a fundamental concept in quantum mechanics that represents the total energy of a system. It has a critical role in quantum systems time evolution and financial markets' dynamics. The main differences between the Hamiltonians of these two models are given below:

First, we introduce the quantum anharmonic oscillator model proposed by Gao and Chen (2017). The Hamiltonian for the quantum anharmonic oscillator can be written as:

$$\hat{H} = -\frac{\hbar^2}{2m} \frac{\partial^2}{\partial r^2} + V(r, t) \quad (1)$$

Here, $-\frac{\hbar^2}{2m} \frac{\partial^2}{\partial r^2}$ represents kinetic energy (KE), $\hbar$ is the reduced Planck's constant, m denotes internal potential (e.g., market capital), and $V(r, t)$ stands for potential energy (PE).

By contrast, the path integral model in quantum finance, introduced

by Baaquie (2007), has a Hamiltonian written as:

$$\hat{H} = -\frac{1}{2m} \frac{\partial^2}{\partial x^2} + V(x) \quad (2)$$

The main difference between the two Hamiltonians lies in potential terms and the meaning of variables involved. In quantum anharmonic oscillator model, the potential term is $V(r, t)$ are interpreted by market participants in Lee's work (Lee, 2020). Meanwhile, for the path integral model, the potential term $V(x)$ depends on the specific financial instrument, without any direct physical interpretation.

These models have different price representations as a result. In quantum anharmonic oscillator model, the asset price is represented by a discrete set of states, each corresponding to a different price. In path integral model, the asset price is represented by a continuous function of time describing the evolution of the asset price over time. Compared to path integral model, the quantum anharmonic oscillator models is simple to solve Quantum Price level.

3.2. Quantum price level

According to Quantum Finance's theories (Lee, 2020), the dynamics of financial instruments (such as currencies, financial indices, cryptocurrencies, etc.) from worldwide markets can be modeled and considered as Financial Particles (FPs) as its main feature is wave-particle duality. FPs' motion and dynamics are subjected to their intrinsic Quantum Energy Fields similar to particles in the physical area. These FPs also have their equilibrium states. When no external stimulus is available, these particles stay in current state. But if external energy is available to stimulate these particles, they become exhilarated and forced to another equilibrium level. These existed levels are called Quantum Price Levels (QPLs) in financial markets and used as technical indicators similar to support and resistance lines. The detailed QPL calculation process is shown below (Lee, 2020):

The first question is to model the potential energy of market movement. Since market actions taken are influenced by its market participants, their behaviors and contributions to the potential energy can be modeled as harmonic oscillator term. As they may cause volatility to the market, a volatility term is defined. There are five major participants in the market:

1. Market Makers
2. Speculators
3. Hedgers
4. Investors
5. Arbitrageurs

Step 1: Model the potential energy of market movement via four major participants. Define z_+ as market demand, and z_- as market supply. The excess demand is given as:

$$\Delta z = z_+ - z_- \quad (3)$$

Indicates instantaneous return as:

$$r(t) = r(t, \Delta t) = \frac{p(t) - p(t - \Delta t)}{p(t - \Delta t)} \quad (4)$$

The relationship between exceed demand and return are denoted as $r(t) = \frac{\Delta z}{\gamma}$, in which γ is a scaling factor represents market depth. Then $\frac{dr}{dt} = \frac{1}{\gamma} \frac{d(\Delta z)}{dt}$ can be derived. So, what to obtain is term $\frac{d(\Delta z)}{dt}$.

For **Market makers**, they work as accommodator in the market absorbing the exceed demand. Define an absorbability factor α , the absorb capacity of the market makers is:

$$\frac{d\Delta z}{dt} \Big|_{MM} = \frac{dz_+}{dt} \Big|_{MM} - \frac{dz_-}{dt} \Big|_{MM} = -\alpha_+ z_+ + \alpha_- z_- \quad (5)$$

For an efficient market, assume $-\alpha_+ = \alpha_-$, and finally the

contribution of the market maker is given as

$$\frac{d\Delta z}{dt} \Big|_{MM} = -\alpha_{MM} \Delta z = -\gamma \alpha_{MM} r \quad (6)$$

where + indicates long position, - represents short position.

For **Speculators**, they seldom think to avoid potential risks. So, their behaviors are regarded as opposition to market. For this reason, a damping term δ to show their resistance to market:

$$\frac{d\Delta z}{dt} \Big|_{SP} = -\delta_{SP} r \quad (7)$$

Hedgers are experienced traders capable to design complex hedging strategies to achieve risk control. They sometimes trade in reverse direction against overall market trends. If using term v to indicate market volatility, hedgers' behaviors are represented by the following equation:

$$\frac{d\Delta z}{dt} \Big|_{HG} = -(\delta_{HG} - v_{HG} r^2) r \quad (8)$$

Investors behave similar to hedgers. They possess fundamental consciousness about stop loss. Their two main characteristics are to: 1) make profits by following trends and 2) risk control. So, the potential energy formula can be given as:

$$\frac{d\Delta z}{dt} \Big|_{IV} = (\delta_{IV} - v_{IV} r^2) r \quad (9)$$

where δ denotes the harmonic dynamic term (trend following contribution), and v represents the anharmonic term (market volatility).

After building these models, the Quantum Dynamics of all key participants are combined as follows:

$$\begin{aligned} \frac{d\Delta z}{dt} &= \frac{d\Delta z}{dt} \Big|_{MM} + \frac{d\Delta z}{dt} \Big|_{SP} + \frac{d\Delta z}{dt} \Big|_{HG} + \frac{d\Delta z}{dt} \Big|_{IV} \\ &= -\gamma \alpha_{MM} r - \delta_{SP} r + (\delta_{HG} - \delta_{HG} r^2) r + (\delta_{IV} - \delta_{IV} r^2) r \\ &= -\delta r + v r^3 \end{aligned} \quad (10)$$

and finally rewrite $\frac{dr}{dt}$ as:

$$\frac{dr}{dt} = \gamma \frac{d(\Delta z)}{dt} = -\gamma \delta r + \gamma v r^3 \quad (11)$$

where $\delta = \gamma \alpha_{MM} + \delta_{SP} + \delta_{HG} - \delta_{IV}$ (damping term) and $v = v_{HG} - v_{IV}$ (volatility term).

Apply $\frac{dr}{dt}$ as the Brownian price return in Langevin equation, the potential energy model is given as:

$$V(r) = \int (-\gamma \eta \delta r + \gamma \eta v r^3) dr = \frac{\gamma \eta \delta}{2} r^2 - \frac{\gamma \eta v}{4} r^4 \quad (12)$$

where η is the market resistance coefficient.

Step 2: Model the kinetic term of market movement via price return. The second question is how to model the kinetic energy of market movement. Since financial particles have wave-particle dualism, it can use Schrodinger equation.

$r(t)$ is used to simulate the displacement of energy. The return changes as time series move so the problem is simplified by using time-independent Schrodinger equation. Then Hamiltonian operator for financial particles are defined as:

$$\hat{H} = -\frac{\hbar}{2m} \frac{\partial}{\partial r^2} + V(r) \quad (13)$$

where $\frac{\hbar}{2m} \frac{\partial}{\partial r^2}$ is kinetic energy (KE), $\hbar$ is the Planck's constant, m stands for internal potential (e.g., market capital), $V(r)$ represents potential energy (PE).

This corresponds to the stationary market environment in Classical Computational Finance Theory (CFT), to analogue such quantum dynamics as time-independent Schrödinger equation given by:



Fig. 1. a-c Three typical positions of QPLs in relation to the daily price movement (based on forex products).

$$\left[\frac{-\hbar}{2m} \frac{d^2}{dr^2} + \left(\frac{\gamma\eta\delta}{2} r^2 - \frac{\gamma\eta\nu}{4} r^4 \right) \right] \varphi(r) = E\varphi(r) \quad (14)$$

where E is the eigen-energy of the financial market and $\varphi(r)$ is the corresponding eigen-function.

Step 3: Perform action space search by solving the QFSE. According to Dasgupta's method to define Quantum Anharmonic Oscillator as:

$$H^m(\lambda)\psi = -\frac{d^2\psi}{dx^2} + (x^2 + \lambda x^{2m})\psi = E\psi \quad (15)$$

where energy term E is estimated by:

$$\left(\frac{E(m,n)}{n+1} \right)^{(m+1)} - \left(\frac{E(m,n)}{n+1} \right)^{(m-1)} = \left(K_0^{(m,n)} \right)^{m+1} \lambda \quad (16)$$

where $E^{(m,n)}$ is the n -level energy, $\left(K_0^{(m,n)} \right)$ is a constant.

Then set $m = 2$:

$$-\frac{d^2\psi_r}{dr^2} + (r^2 + \lambda r^4)\psi = E\psi_r \quad (17)$$

Also, we have:

$$\left(\frac{E(n)}{n+1} \right)^3 - \left(\frac{E(n)}{n+1} \right) = (K_0(n))^{3+1} \lambda \quad (18)$$

Since $K_0(n) = \left[\frac{1.1924+33.2383n+56.2169n^2}{1+43.6196n} \right]^{\frac{1}{3}}$. To solve the equation, the value of λ is necessary. Using finite difference method, derive:

$$\lambda = \frac{r_{-1}^2\psi_{-1} - r_{+1}^2\psi_{+1}}{r_{+1}^4\psi_{+1} - r_{-1}^4\psi_{-1}} \quad (19)$$

where ψ is the wavefunction indicating the probability distribution of the historical price return, r_{-1} and r_{+1} are adjacent to the r_0 , and r_0 stands for the return with maximum probability.

Finally, use Gerolamo Cardano's method to solve the depressed cubic equation:

$$E(n) = \sqrt[3]{-\frac{q}{2} + \sqrt{\frac{q^2}{4} + \frac{p^3}{27}}} + \sqrt[3]{-\frac{q}{2} - \sqrt{\frac{q^2}{4} + \frac{p^3}{27}}} \quad (20)$$

where $p = -(2n+1)^2 q = -\lambda(2n+1)^3 [K_0(n)]^3$

Then evaluate $E(n)$ to obtain actual QPLs values.

Algorithm 1 QPL calculation Algorithm.

1. For each financial product **do**
2. Read Daily time series and extract (Date, OPEN, HIGH, LOW and CLOSE)
3. Calculate daily price return $r(t)$
4. Evaluate quantum price return wave function $\psi(r)$
5. Calculate λ using F.D.M
6. Calculate σ , the standard deviation of $\psi(r)$
7. Obtain normalized max quantum price return ($maxNQPR$) from $\psi(r)$

(continued on next column)

(continued)

8. Solve QFSE by Gerolamo Cardano method to calculate first 21 Quantum Finance Energy level ($QFEL_0, \dots, QFEL_{21}$)
9. Calculate normalized quantum price return for QPLs: $NQPR_i = 1 + 0.21 \cdot \sigma \cdot \frac{QFEL_i}{QFEL_0}$
10. End For

3.3. QPL-inspired risk control mechanism

An important characteristic of QPL is that it models the equilibrium states of price movement of a financial product on daily basis as mentioned in Section 3.2.

In other words, QPL is a robust and important sign that indicates oscillating prices support and resistance levels to estimate the potential price movement intervals, and make trading decisions accordingly.

Fig. 1 above shows 3 specific situations during the training period. Here three types of trading decisions are explained: 1) Base on QPL_{+1} 2) Base on QPL_{+2} 3) Hold the order.

1. In Fig. 1a), the interval $[QPL_{+1}, QPL_{+2}]$ is a subset constituted by that day's highest price and lowest price which is $[LOW, HIGH]$. If QPL_{+1} is selected by agent as the key factor for trading decision making, the agent is to be guided to close the holding order of the selected product at QPL_{+1} . Similarly, if QPL_{+2} is selected, agent will close the order at exactly QPL_{+2} . Besides, if the agent decides to do nothing in a specific period, then the pending order of this product is held.
2. In Fig. 1b), not both QPL_{+1} and QPL_{+2} are within the interval $[LOW, HIGH]$. In this case, the operation after making trading decision based on QPL_{+1} is the same as in Fig. 1a). But once QPL_{+2} is selected as the key factor to make trading decision, the agent is going to be trained to hold the current order.
3. In Fig. 1c), neither QPL_{+1} nor QPL_{+2} are within the interval constituted by $[LOW, HIGH]$. In this case, the only operation required is to hold the order regardless of which QPL is selected by agent to make the trading decision.

3.4. Problem definition and assumption

Portfolio management in a given period means a forex trader can invest into a set of financial assets and reallocate one's capital, for profit maximization. The following assumptions are:

- the trading volume of financial products selected for experiments in simulated market assumed to be sufficient, hence all trading decisions made by the agent are insignificant to affect the overall market.
- no slippage occurred in the market. As a highly liquidated market environment is simulated for the experiment, each trading operation execution is immediate and processed at precise prices when these orders are opened.

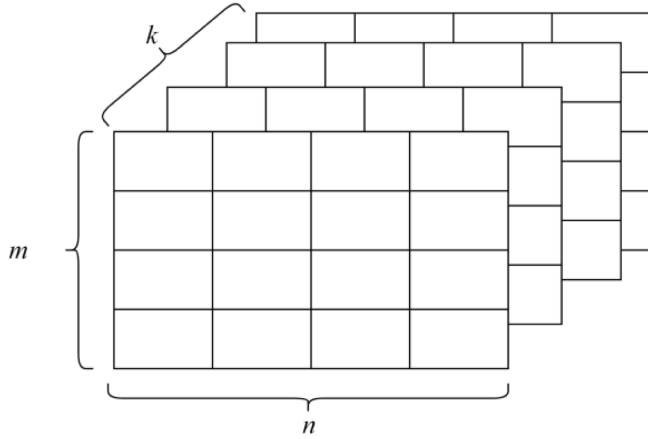


Fig. 2. The representation of state space.

- Different exchanges have different trading rules. We have assumed that our virtual environment supports T0 trading, which means that a trader can buy and sell the same financial products within the same day. Under this assumption, the simulated trading mechanism in the virtual environment can be described as follows: As the market commences trading in T0, the agent opens positions according to the portfolio weights generated by the agent. Throughout the trading period, the risk control mechanism can close orders as mentioned in Section 3.3. As the trading period ends at T0, all positions for all financial products will be closed. This trading mechanism is followed by the virtual environment every trading day.

3.5. Mathematical formalism

If $m+1$ assets are selected in the portfolio, their close prices can constitute a vector at period t , represented by $v_t = (v_{0,t}^{close}, v_{1,t}^{close}, \dots, v_{m,t}^{close})^T$. $v_{i,t}$ represents the i th asset's close price at period t . Quoted currency is a special item denoted as $v_{0,t}^{close}$ in the portfolio, which corresponds to balance amount remained for purchasing other financial products. Quoted currency is a risk-free asset compared with other m forex products. So, its value can be set to be always 1.

Additionally, the relative price vector in t^{th} trading period can be defined as y_t . It is calculated by element-wise division of v_{t+1} by v_t or the value of selected QPL divided by v_t :

$$y_t = \begin{cases} \left(1, \frac{v_{1,t+1}^{close}}{v_{1,t}^{close}}, \frac{v_{2,t+1}^{close}}{v_{2,t}^{close}}, \dots, \frac{v_{m,t+1}^{close}}{v_{m,t}^{close}}\right)^T, & \text{when } QPL_i \notin [v_{m,t+1}^{low}, v_{m,t+1}^{high}] \\ \left(1, \frac{v_{1,t+1}^{QPL_i}}{v_{1,t}^{QPL_i}}, \frac{v_{2,t+1}^{QPL_i}}{v_{2,t}^{QPL_i}}, \dots, \frac{v_{m,t+1}^{QPL_i}}{v_{m,t}^{QPL_i}}\right)^T, & \text{when } QPL_i \in [v_{m,t+1}^{low}, v_{m,t+1}^{high}] \end{cases} \quad (21)$$

where i stands for an integer indicating to a particular QPL selected.

Each element of y_t represents the quotient of closing prices of individual asset in period t . If defining the value of whole portfolio at the end of period t as p_t , according to the assumption made above, p_{t-1} can be used to denote the portfolio value at the start of period t . So, the price relative vector can be used to compute the cumulative return R at time t as:

$$R_t = \frac{p_t}{p_0} \quad (22)$$

and the return r at time t is given as:

$$r_t = \frac{p_t}{p_{t-1}} \quad (23)$$

Without considering the transaction commission, the return gained at time t is given as:

$$r_t = w_{t-1} \cdot y_t \quad (24)$$

where vector w_t represents the portfolio weight at the end of period t (also the beginning of period $t+1$). $w_{t,i}$ means the percentage of i th asset in the portfolio after capital allocation adjustments. Its initial state w_0 should be $w_0 = (1, 0, 0, \dots, 0)^T$. There is a rule existed for all the weight elements:

$$\sum_i w_{t,i} = 1, \forall t \quad (25)$$

Therefore, the return accumulated from 0 to T can be expressed as:

$$R_T = \prod_{t=1}^T r_t = \prod_{t=1}^T w_{t-1} \cdot y_t \quad (26)$$

3.6. Transaction cost

All buy or sell trading operations require commission fee in real-world market. Algorithm dynamics can only allocate capital to maximize the highest expected increase rate without transaction cost. The transaction costs may outweigh returns, which means the corresponding actions used to be optimal are not valid. Therefore, it is necessary to introduce transaction cost so that QFPIS is formulated to market reality.

At the start of period t , the portfolio's weighting is denoted as w_{t-1} . Because price movements occur continuously in market, the vector representing portfolio weighting can transform correspondingly into the following expression (Ormos & Urbán, 2013):

$$w'_t = \frac{y_t \odot w_{t-1}}{y_t \cdot w_{t-1}} \quad (27)$$

where $\odot$ denotes element-wise multiplication.

Similarly, at the begin of period t , the portfolio value is defined as p_{t-1} , as stated by Jiang, Xu, and Liang (2017), after the price movement, a transaction remainder factor μ_t is defined in the training process and the portfolio value p'_t at the end, the portfolio value p_t at $t+1$ is evolved to:

$$p_t = \mu_t p'_t$$

For training environment, when the transaction cost of buying and selling is the same (i.e., $c_p = c_s = c$), μ_t is approximated as:

$$\mu_t = \left(1 - C \sum_{i=1}^m |w_{i,t} - w'_{i,t}|\right) \quad (28)$$

In the back-test environment, μ_t is converged according to the formula below according to (Jiang et al., 2017):

$$\mu_t = \frac{1}{1 - c_p w_{t,0}} \left(1 - c_p w'_{t,0} - (c_s + c_p - c_s c_p) \sum_{i=1}^m (w'_{t,i} - \mu_t w_{t,i})\right) \quad (29)$$

After obtaining the expression of transaction cost, the return rate at time t and cumulative return from 0 to T is re-calculated as:

$$r_t = \mu_t w_{t-1} \cdot y_t \quad (30)$$

$$R_T = \prod_{t=1}^T r_t = \prod_{t=1}^T \mu_t w_{t-1} \cdot y_t \quad (31)$$

Finally, the reward function is defined as:

$$r_t := \ln r_t = \ln(\mu_t w_{t-1} \cdot y_t) \quad (32)$$

There is a possible defect of the current reward function is that during bullish/bearish phase, it may always provide positive/negative feedback regardless of agent's actions. A possible way to address this problem is to add a new term to the reward function to improve agent's actions judgement. This term represents the ratio of portfolio weight vector to daily price change. With this addition, during bullish/bearish phase, the agent's actions receive more/less rewards/punishment if they

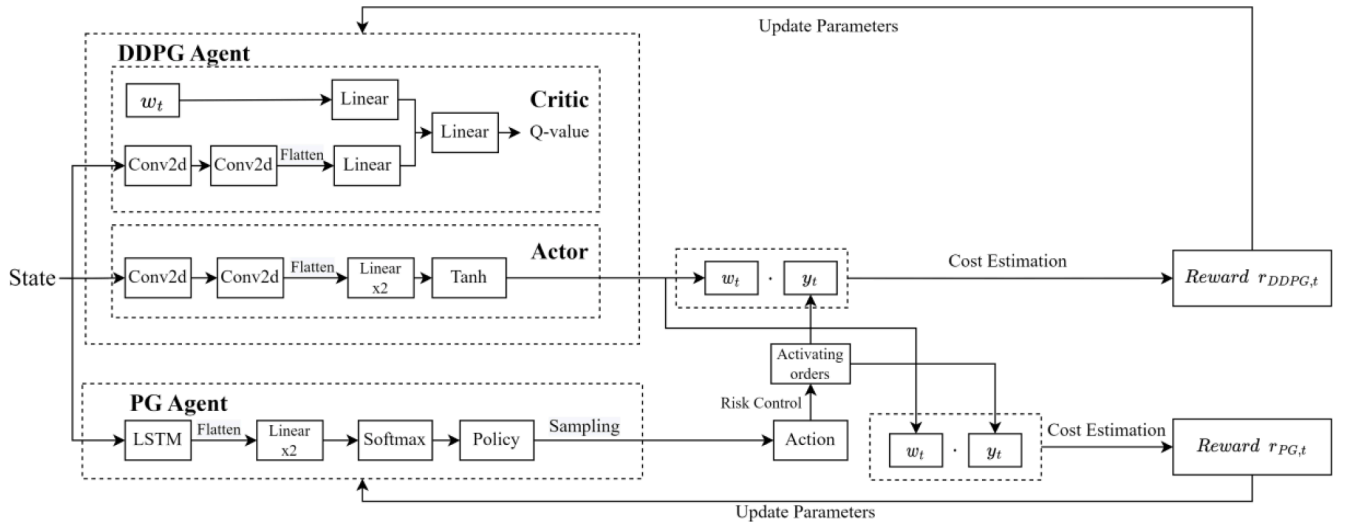


Fig. 3. The Overall Architecture of Quantum Finance Portfolio Investment System.

Table 1
Experiment Dataset Information.

Market	Symbol	Train Date	Val Date	Test Date
Forex Market	AUDCAD, AUDUSD, EURAUD, EURCAD, EURUSD, GBPUSD, NZDCHF, NZDUSD, USDCHF	2006/05/16–2019/12/04 (Length:3501 trading days)	2019/12/06–2021/08/09 (Length:438 trading days)	2021/08/11–2023/04/13 (Length: 438 trading days)
The U.S. stock	AAPL, GE, IBM, JNJ, JPM, KO, MSFT, V, XOM	2008/03/19–2020/04/07 (Length:3035 trading days)	2020/04/09–2021/10/07 (Length:379 trading days)	2021/10/11–2023/04/13 (Length: 379 trading days)
China A-share market	000002.SZ, 002035.SZ, 600036.SH, 600019.SH, 601318.SH, 601988.SH, 600519.SH, 600028.SH, 600276.SH	2007/03/01–2020/06/11 (Length: 2769 trading days)	2020/06/15–2021/11/15 (Length: 346 trading days)	2021/11/17–2023/04/19 (Length: 346 trading days)

distribute more/less weights to the product with a larger price rise/fall.

3.7. The application of deep reinforcement learning techniques

3.7.1. Definition of state space, action space and reward function

State: The state space of QFPIS model is a three-dimensional matrix with a shape of (k, m, n) , where k denotes the number of attributes, m represents the quantity of pre-selected financial products in the portfolio, and n signifies the size of the sliding window. Fig. 2 illustrates the state space representation.

The state space is constructed by five attributes, namely: 1) Daily Price Return, 2) Relative Strength Index (RSI), 3) Moving Average Convergence/Divergence (MACD), 4) Stochastic Oscillator Indicator (KDJ), and 5) William %R (WILLR).

Action: Each action is defined as period t 's desired weighting vector, denoted as:

$$w_t = (w_{0,t}, w_{1,t}, w_{2,t}, \dots, w_{m,t})^T \quad (33)$$

Reward: Considering the transaction cost, logarithmic rate of return is used in RL as the reward function following (Jiang, Xu, & Liang, 2017):

$$r_t = \ln \frac{p_t}{p_{t-1}} = \ln(\mu_t y_t \cdot w_{t-1}) \quad (34)$$

3.7.1.1. Deep deterministic policy gradient. In Policy Gradient, the main idea is to optimize the policy represented by Deep Neural Network parameters. Each iteration is required to sample trajectories with size N $\{\tau_i\}_{i=1}^N$ to update gradients. However, in some complex situations, Q-Learning Algorithm is not appropriate due to 2 main reasons. First, the action space may be too large to be represented by a discrete Q-table. Second, determining the global optimal policy in an arbitrary Q-value function may be infeasible if some good features guaranteed such as

convex are available. (Hafner & Riedmiller, 2011). Therefore, based on Actor-Critic Framework in traditional RL and idea of DQN, Lillicrap et al (Schulman et al., 2016) transformed Deterministic Policy Gradient (DPG) Algorithm into Deep Deterministic Policy Gradient (DDPG) Algorithm.

In DDPG, actor is responsible for outputting continuous action and critic would perform evaluation and policy improvement on which is a Q-function approximator to represent target function. Parameterized by θ , it can be expressed by:

$$\tau = (s_1, a_1, s_2, a_2, \dots)$$

$$J(\pi_\theta) = \mathbb{E}_{\tau \sim p_\theta(\tau)} \left[\sum_t \gamma^t r(s_t, a_t) \right]$$

$$\pi_{\theta^*} = \underset{\pi_\theta}{\operatorname{argmax}} J(\pi_\theta)$$

$$= \underset{\pi_\theta}{\operatorname{argmax}} \mathbb{E}_{\tau \sim p_\theta(\tau)} \left[\sum_t \gamma^t r(s_t, a_t) \right]$$

$$= \underset{\pi_\theta}{\operatorname{argmax}} \mathbb{E}_{\tau \sim p_\theta(\tau)} [r(\tau)] = \underset{\pi_\theta}{\operatorname{argmax}} \int \pi_\theta(\tau) r(\tau) d\tau \quad (35)$$

Assume the time horizon is finite, the derivative of policy can be:

$$\pi_\theta(\tau) = \pi_\theta(s_1, a_1, \dots, s_T, a_T)$$

$$= p(s_1) \prod_{t=1}^T \pi_\theta(a_t | s_t) p(s_{t+1} | s_t, a_t) \quad (36)$$

For easy computation, the equation above is transformed into summation forms:

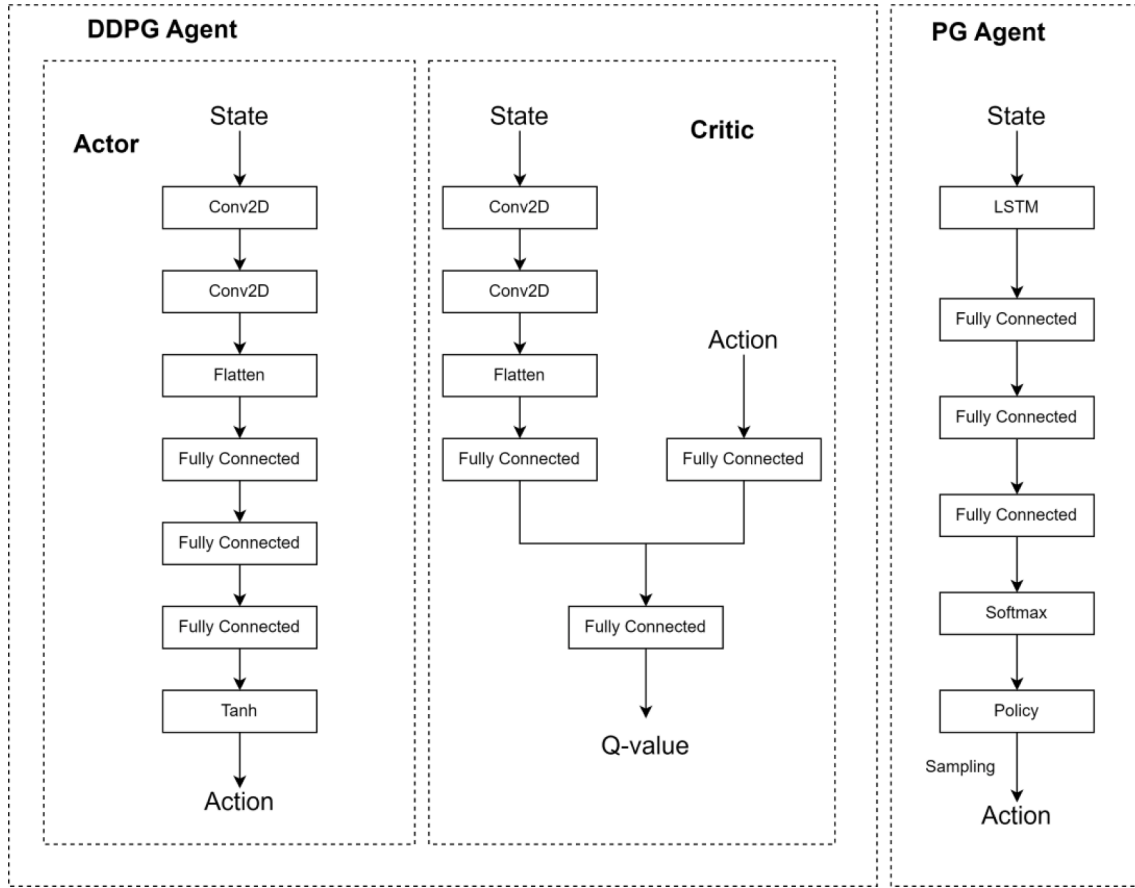


Fig. 4. Structures of DDPG and PG Agents.

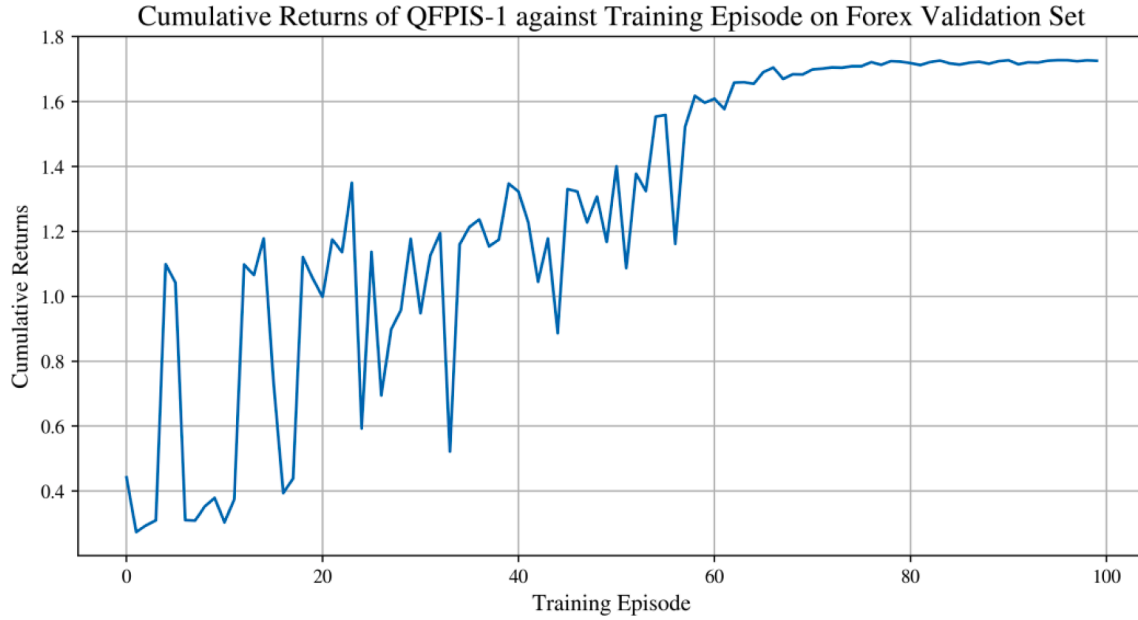


Fig. 5. Cumulative Returns of QFPIS-1 against Training Episode on Forex Validation Set.

$$\begin{aligned}\nabla_{\theta}\pi_{\theta}(\tau) &= \pi_{\theta}(\tau) \frac{\nabla_{\theta}\pi_{\theta}(\tau)}{\pi_{\theta}(\tau)} \\ &= \pi_{\theta}(\tau) \nabla_{\theta} \log \pi_{\theta}(\tau)\end{aligned}$$

$$\nabla_{\theta} \log \pi_{\theta}(\tau) = \nabla_{\theta} \left(\log p(s_1) + \sum_{t=1}^T \log \pi_{\theta}(a_t | s_t) + \log p(s_{t+1}) \right)$$

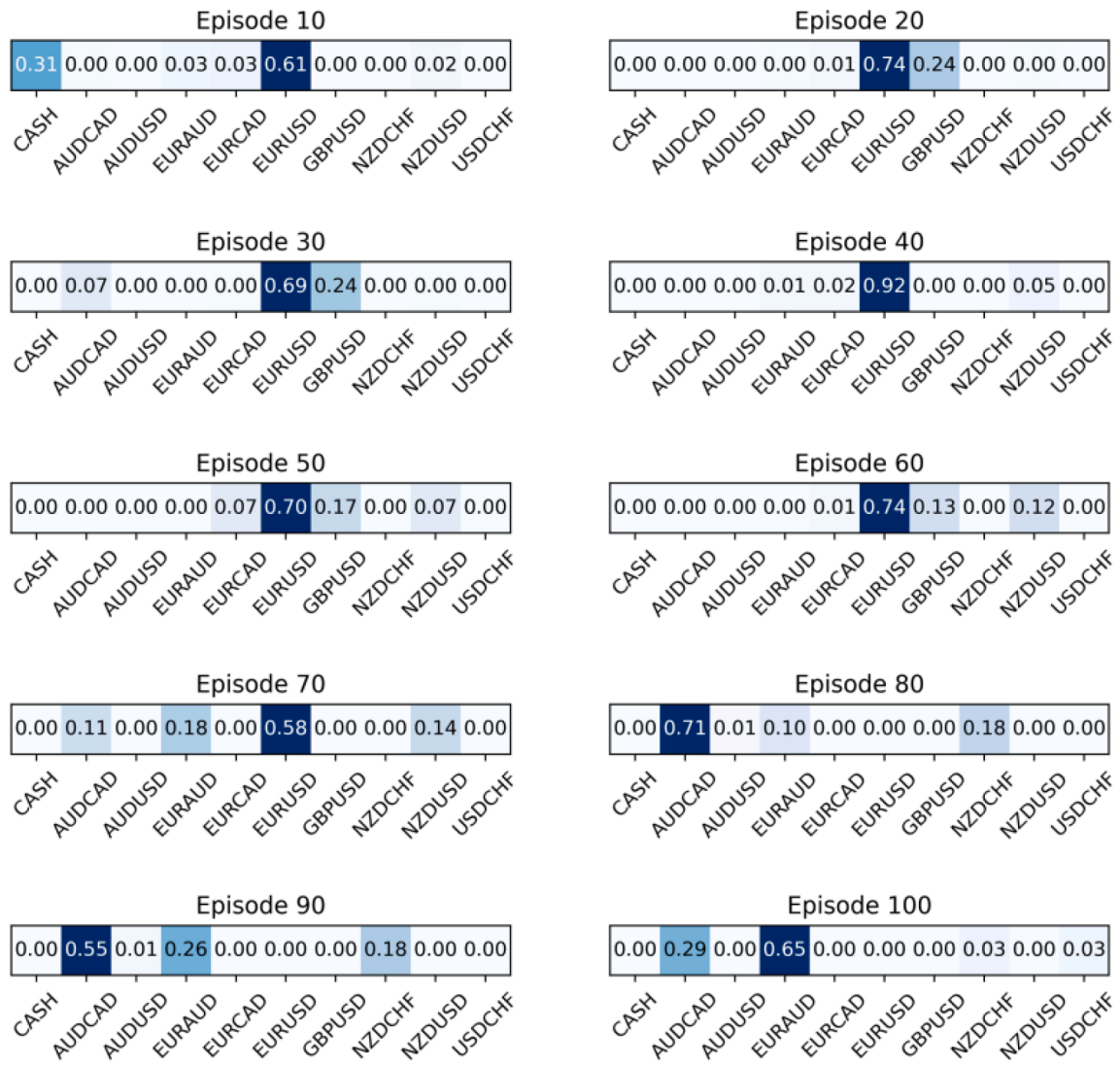


Fig. 6. Cumulative Returns of QFPIS-1 against Training Episode on Forex Validation Set.

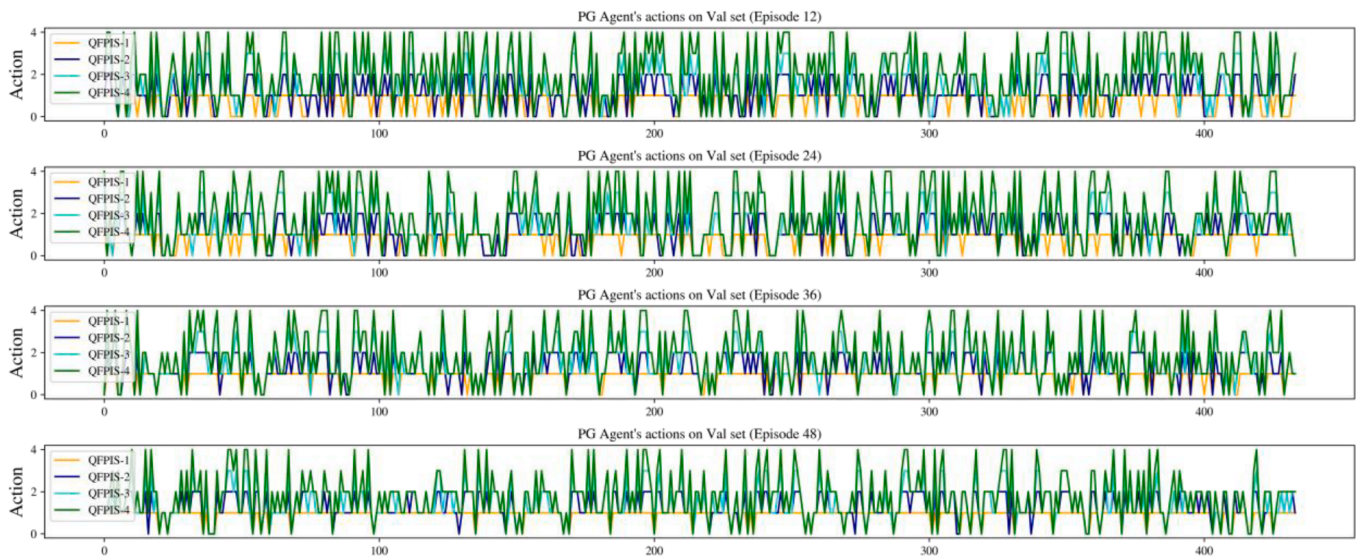


Fig. 7. Cumulative Returns of QFPIS-1 against Training Episode on Forex Validation Set.

Table 2
Performance Metrics on Forex Data.

Model	AR	CRR	SR	MDD	fAPV
QFPIS-1	30.87 %	59.10 %	2.03	0.11 %	1.589
QFPIS-2	39.91 %	78.80 %	1.30	0.09 %	1.783
QFPIS-3	42.54 %	84.64 %	1.29	0.09 %	1.841
QFPIS-4	59.80 %	124.80 %	1.29	0.11 %	2.242
DDPG	-0.23 %	-0.13 %	-0.65	0.45 %	0.996
A2C	-1.33 %	-2.05 %	-1.15	2.08 %	0.977
SAC	-54.10 %	-73.77 %	-38.34	73.77 %	0.262
PPO	-25.91 %	-40.20 %	-21.90	40.20 %	0.597
OLMAR	-1.09 %	-1.64 %	-0.99	1.70 %	0.981
PAMR	-1.17 %	-1.76 %	-1.51	1.86 %	0.980
CWMR	-1.85 %	-3.17 %	-2.07	3.17 %	0.968
RMR	-1.15 %	-1.98 %	-0.98	2.03 %	0.980

Table 3
Performance Metrics on The U.S. Stock.

Model	AR	CRR	SR	MDD	fAPV
QFPIS-1	14.27 %	22.22 %	0.74	25.29 %	1.22
QFPIS-2	14.84 %	23.46 %	0.74	25.28 %	1.23
QFPIS-3	-4.76 %	-7.64 %	-0.30	15.77 %	0.93
QFPIS-4	21.83 %	35.25 %	0.93	27.05 %	1.34
DDPG	-7.56 %	-11.50 %	-0.57	21.94 %	0.89
A2C	15.62 %	23.71 %	1.10	9.14 %	1.24
SAC	-46.78 %	-61.08 %	-3.82	61.55 %	0.39
PPO	-14.68 %	-21.47 %	-0.76	26.90 %	0.79
OLMAR	16.48 %	25.89 %	1.07	9.34 %	1.26
PAMR	-0.55 %	-0.56 %	0.03	13.54 %	0.99
CWMR	13.78 %	21.25 %	0.97	9.99 %	1.21
RMR	17.44 %	27.10 %	1.12	10.03 %	1.27

Table 4
Performance Metrics on China A-share market.

Model	AR	CRR	SR	MDD	fAPV
QFPIS-1	-4.23 %	-5.68 %	-2.02	6.03 %	0.94
QFPIS-2	30.07 %	42.80 %	1.13	18.67 %	1.43
QFPIS-3	-9.12 %	-13.05 %	-0.34	25.24 %	0.88
QFPIS-4	-7.54 %	-10.10 %	-2.39	10.54 %	0.90
DDPG	12.49 %	21.83 %	0.51	27.39 %	1.17
A2C	15.57 %	22.83 %	1.06	14.17 %	1.22
SAC	-44.81 %	-55.07 %	-3.31	56.94 %	0.45
PPO	-41.72 %	-51.36 %	-2.67	53.02 %	0.48
OLMAR	15.38 %	19.74 %	0.99	15.60 %	1.22
PAMR	8.93 %	10.72 %	0.74	13.94 %	1.12
CWMR	10.84 %	15.03 %	0.82	13.69 %	1.15
RMR	14.32 %	19.98 %	0.93	15.86 %	1.20

$$= \sum_{t=1}^T \nabla_{\theta} \log \pi_{\theta}(a_t, s_t) \quad (37)$$

Hence, rewrite the differentiation of target function into logarithm of policy:

$$\begin{aligned} \nabla J(\pi_{\theta}) \mathbb{E} &= \mathbb{E}_{\tau \sim p_{\theta}(\tau)} [r(\tau)] \\ &= \mathbb{E}_{\tau \sim p_{\theta}(\tau)} [\nabla_{\theta} \log \pi_{\theta}(\tau) r(\tau)] \\ &= \mathbb{E}_{\tau \sim p_{\theta}(\tau)} \left[\left(\sum_{t=1}^T \nabla_{\theta} \log \pi_{\theta}(a_t | s_t) \right) \left(\sum_{t=1}^T \gamma^t r(s_t, a_t) \right) \right] \end{aligned} \quad (38)$$

In DDPG, there are 4 networks:

$$\text{Actor}(\mu \text{function}) \begin{cases} \text{online} : \mu(s | \theta^{\mu}) : \text{update} \theta^{\mu} \text{ by gradient} \\ \text{target} : \mu(s | \theta^{\mu'}) : \text{softly update} \theta^{\mu'} \end{cases} \quad (39)$$

$$\text{Critic}(Q \text{function}) \begin{cases} \text{online} : Q(s, a | \theta^Q) : \text{update} \theta^Q \text{ by gradient} \\ \text{target} : Q(s, a | \theta^Q) : \text{softly update} \theta^Q \end{cases} \quad (40)$$

After an agent completed an observation of current state, online actor will decide the “optimal” action based on its policy and state observed. Then online critic will estimate actor’s proposed action, and update online actor using policy gradient. Online critic is then updated according to parameters of target critic and target actor. Finally, target actor and target critic are softly updated according to online critic and online actor.

DDPG update mechanism is as below:

For online actor:

$$\begin{aligned} \nabla_{\theta^{\mu}} J &\approx \mathbb{E}_{s_t \sim \rho^{\theta}} [\nabla_{\theta^{\mu}} Q(s, a | \theta^Q) | s = s_t, a = \mu(s_t | \theta^{\mu})] \\ &= \mathbb{E}_{s_t \sim \rho^{\theta}} \left[\nabla_a Q(s, a | \theta^Q) \Big|_{s=s_t, a=\mu(s_t)} \nabla_{\theta^{\mu}} \mu(s | \theta^{\mu}) \Big|_{s=s_t} \right] \end{aligned} \quad (41)$$

For online critic, the updating process is similar to online actor: Minimize the loss,

$$L = \frac{1}{N} \sum_i (y_i - Q(s_i, a_i | \theta^Q))^2 \quad (42)$$

$$y_i = r_i + \gamma Q'(s_{i+1}, \mu'(s_{i+1} | \theta^{\mu'}) | \theta^Q)$$

For target networks:

$$\theta^Q \leftarrow \tau \theta^Q + (1 - \tau) \theta^Q \quad (43)$$

$$\theta^{\mu} \leftarrow \tau \theta^{\mu} + (1 - \tau) \theta^{\mu} \quad (44)$$

According to the outcomes of DeepMind’s (Lillicrap et al., 2016), the detailed DDPG algorithm is as follows:

Algorithm 2 Deep Deterministic Policy Gradient Algorithm.

1. Initialize critic $Q(s, a | \theta^Q)$ and actor $\mu(s | \theta^{\mu})$ randomly
2. Initialize Q' with parameter $\theta^Q \leftarrow \theta^Q$ and μ' with parameter $\theta^{\mu} \leftarrow \theta^{\mu}$
3. Initialize Experience Replay Buffer R
4. **For** episode = 1 to M **do**
5. Initialize a random process $\mathcal{N}$ for action selection
6. Receive initial observation state s_1
7. **For** $t = 1$ to T **do**
8. Select action $a_t = \mu(s_t | \theta^{\mu}) + \mathcal{N}_t$
9. Execute action a_t , then observe reward r_t and next state s_{t+1}
10. Save the transition (s_t, a_t, r_t, s_{t+1}) into experience replay buffer R
11. Sample a random mini-batch from R with N transitions (s_i, a_i, r_i, s_{i+1})
12. Set y_i value to be $y_i = r_i + \gamma Q'(s_{i+1}, \mu'(s_{i+1} | \theta^{\mu'}) | \theta^Q)$
13. Update critic by minimizing the loss $L = \frac{1}{N} \sum_i (y_i - Q(s_i, a_i | \theta^Q))^2$
14. Update actor policy by sampled policy gradient
15. $\nabla_{\theta^{\mu}} J \approx \frac{1}{N} \sum_i \nabla_a Q(s, a | \theta^Q) \Big|_{s=s_i, a=\mu(s_i)} \nabla_{\theta^{\mu}} \mu(s_i | \theta^{\mu}) \Big|_{s_i}$
16. Update 2 target networks gradually:
17. $\theta^Q \leftarrow \tau \theta^Q + (1 - \tau) \theta^Q$
17. $\theta^{\mu} \leftarrow \tau \theta^{\mu} + (1 - \tau) \theta^{\mu}$
17. **End For**
17. **End For**

3.7.1.2. Policy gradient. The QPL-inspired risk control mechanism shown in section 3.3 with 3 kinds of actions can be taken. The algorithm implemented this mechanism is Policy Gradient (PG) (Silver et al., 2014) (Thomas & Brunskill, 2000). According to PG theories, the policy is updated directly. If denoting a trajectory $\tau = s_1, a_1, s_2, \dots$, the possibility for each trajectory is computed through the following expression:

$$\begin{aligned} p_{\theta}(\tau) &= p(s_1) p_{\theta}(a_1 | s_1) p(s_2 | s_1, a_1) p_{\theta}(a_2 | s_2) \dots \\ &= p(s_1) \prod_{t=1}^T p_{\theta}(a_t | s_t) p(s_{t+1} | s_t, a_t) \end{aligned} \quad (45)$$

After an action is taken by the agent, the environment will randomly return a state. Hence, for PG algorithm, its primary goal is to optimize the value of desired accumulative reward, given as:

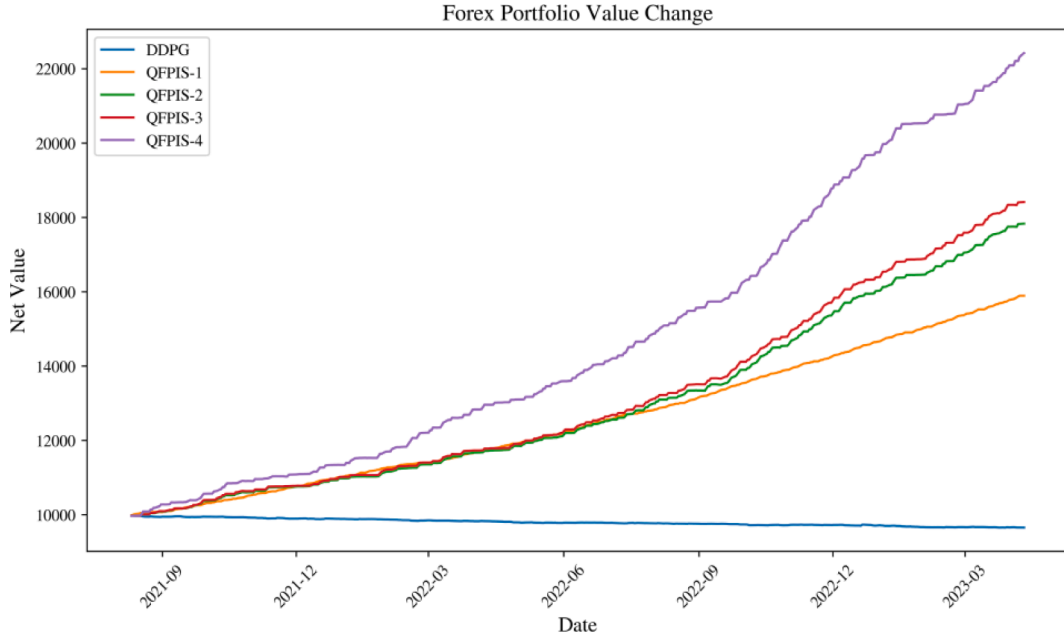


Fig. 8. Portfolio values change during the testing period on Forex dataset.

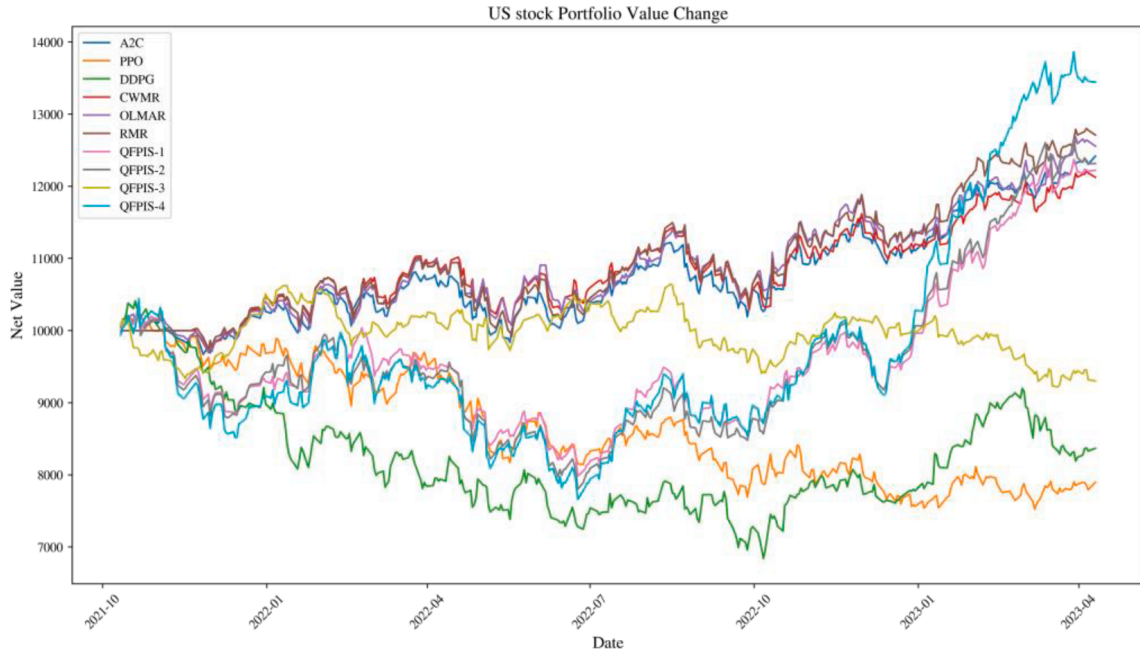


Fig. 9. Portfolio values change during the testing period on the U.S. stock.

$$\bar{R}_\theta = \sum_{\tau} R(\tau) p_\theta(\tau) = E_{\tau \sim p_\theta(\tau)} [R(\tau)] \quad (46)$$

Use a_t to represent the action at time t in the proposed model, to obtain the following expression:

$$a_t = (a_{hold}, a_{QPL_{+1}}, a_{QPL_{+2}}) \quad (47)$$

where three elements in vector a_t represents probability that the decision are made by the agent. By selecting option with the largest probability from these decisions, agent's action is selected:

$$d_t = \operatorname{argmax}(a_t) \quad (48)$$

4. System implementation and experiments

4.1. System architecture

QFPIS is comprised of two major components:

- DDPG Agent
- PG Agent

The overall structure of Quantum Finance Portfolio Investment System is shown in Fig. 3:

After receiving the state at period s_t , DDPG agent can generate the action at period t $a_t = w_t$, and the expected weighting vector is available

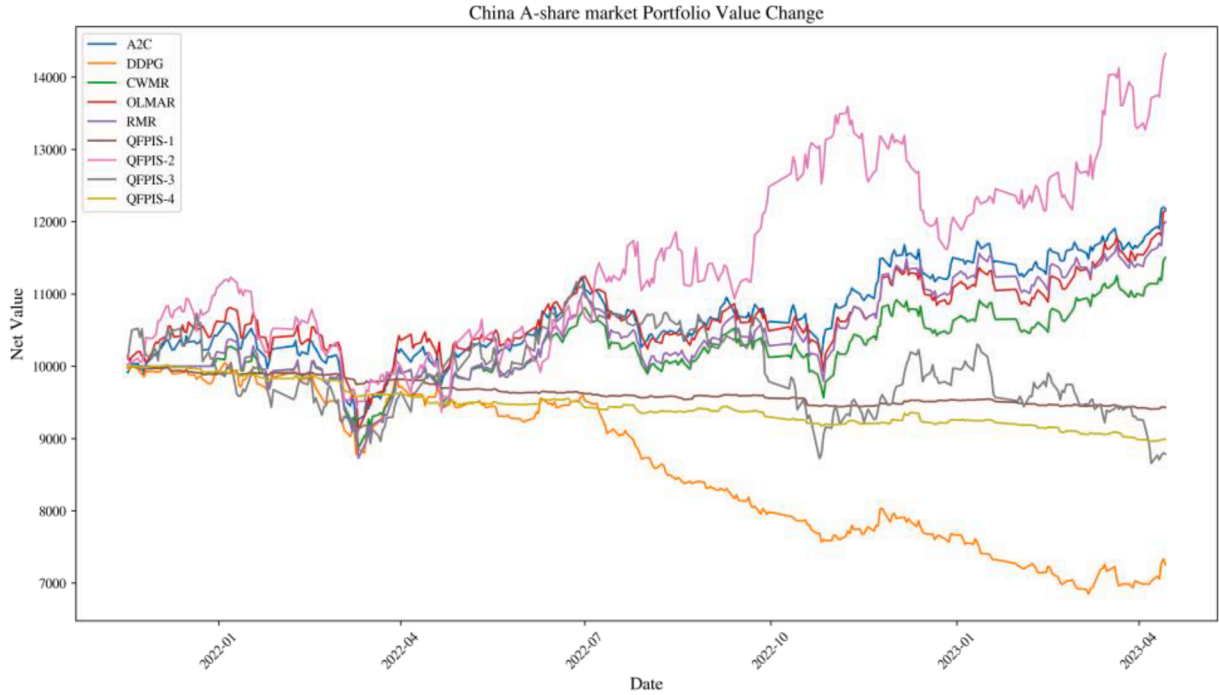


Fig. 10. Portfolio values change during the testing period on the China A-share market.

Table 5

Ablation Study of DDPG and A2C on Forex market.

Model	CRR	MDD	SR	CRR impact	MDD impact	SR impact
NOQPL(DDPG)	-0.13 %	0.455 %	-0.65	-	-	-
+QPL1	59.10 %	0.106 %	2.03	+59.22 %	-0.349 %	+2.68
+QPL2	78.80 %	0.092 %	1.30	+78.93 %	-0.363 %	+1.95
+QPL3	84.64 %	0.092 %	1.29	+84.76 %	-0.363 %	+1.94
+QPL4	124.80 %	0.106 %	1.29	+124.93 %	-0.349 %	+1.94
NOQPL(A2C)	-2.05 %	2.076 %	-1.15	-	-	-
+QPL1	45.37 %	1.385 %	2.01	+47.42 %	-0.690 %	+3.16
+QPL2	61.00 %	1.414 %	1.30	+63.05 %	-0.661 %	+2.45
+QPL3	58.44 %	1.414 %	1.03	+60.49 %	-0.661 %	+2.18
+QPL4	57.73 %	1.411 %	1.01	+59.79 %	-0.664 %	+2.16

to serve for portfolio adjustment at time t . After PG agent decides on whether to hold the activating orders or close them according to a specific QPL, it will define the weighting vector w_t generated by DDPG agent. In the phase of cost estimation, w_t is necessary to calculate of return $r_{DDPG,t}$ and $r_{PG,t}$ for both agents, respectively. Based on a pre-defined commission rate constant C , cost generated in transactions are assessed and the accumulated reward is computed accordingly. After both agents' parameters are updated, the entire iteration is completed.

4.2. Experimental setting

Three portfolios and their dataset information, each from a different market are listed in Table 1:

All data are obtained by the Yahoo Finance API. During the implementation phase, first 80 % of the data is used as the training set, while the subsequent 10 % and the remaining 10 % were used as validation and testing sets respectively. The dates of the training, validation, and testing sets for each dataset are listed in Table 1. A sliding window with a size of $n = 3$ is used for the state space. Additionally, in the back-testing stage, a series of initial parameters are used, including a portfolio value $p_0 = 10000$ (USD) and a commission rate $C = 0.25\%$. Two measures are employed to prevent overfitting:

- 1) Use learning rate warmup and cosine learning rate decay in Transformer (Ashish Vaswani et al., 2017).
- 2) The validation set is used for early stopping during training. The best model is saved based on the highest final Accumulated Portfolio Value in validation set.

4.3. Network structure and training setting

CNN outperformed than RNN represented by LSTM for the actor and critic network implementation in DDPG algorithm according to results by researchers (Jiang et al., 2017). Therefore, a two-layer CNN structure is used in both networks of DDPG agent as shown in Fig. 4. For actor network, its remaining structure comprised of three FC (fully connected linear) layers with a Tanh layer. For critic network, although CNN structure part is same as actor network counterpart, but a channel dedicated for the input of action space followed by a fully connected layer is added. Consequently, the outputs from these two branches are integrated and imported to a fully connected layer. After all the structures are formed, Q-value is generated. In PG agent design, since LSTM is effective and efficient at sequential data processing (Gers, Schmidhuber, & Cummins, 2000), observations are first imported into LSTM network for feature extraction, then two FC layers and a SoftMax layer are constructed to receive and process output from LSTM layer, subsequently an

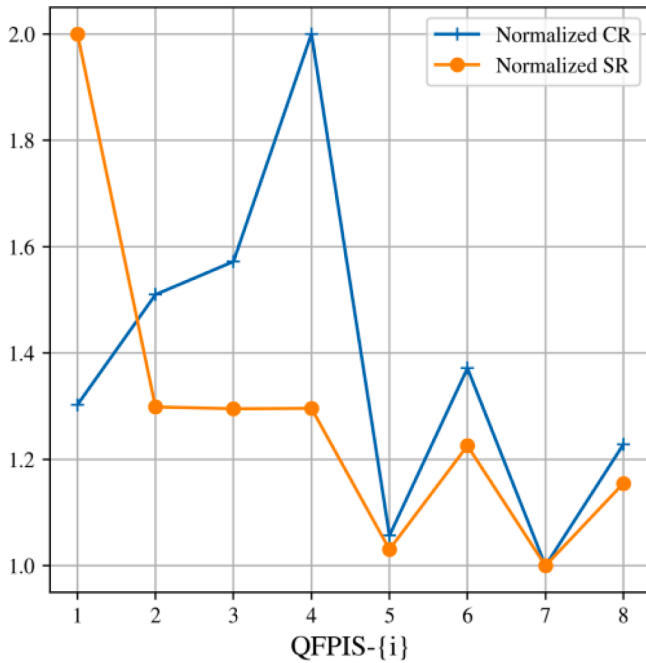


Fig. 11. Risk control action space's effect on CR and SR.

action is sampled from generated policy. The structures of DDPG and PG Agents are illustrated in Fig. 4.

4.4. Parameter settings

There were 100 epochs required for both agents in the training process while the optimizer used for gradient descent was Adaptive Moment Estimation (ADAM). Regarding the learning rates of DDPG agent, the actor network was assigned to 0.0001, while the critic network was set to 0.001. It is noted that the learning rate for PG network is 0.0001. These three networks have their learning rates warmed up linearly over 10 steps and subsequently decayed following a cosine annealing strategy. The replay buffer size was set to 10,000 and batch size was 64. The discounting factor γ was configured to 0.99. The parameter τ used for soft updating is 0.001. Ornstein-Uhlenbeck Noise (Lillicrap et al., 2016) is employed in the action selection process to encourage exploration of the environment, with the parameters θ and σ for the noise set as 0.15 and 0.01, respectively. The training parameters are the same for all three markets.

5. System performance

5.1. Model setting

To compare performance between QFPIS and related models, four mainstream online portfolio selection strategies (Li & Hoi, 2013) and three models implemented through reinforcement learning algorithms are selected.

Online portfolio selection strategies:

- OLMAR (On-Line Portfolio Selection with Moving Average Reversion) (Li & Hoi, 2012)
- PAMR (Passive Aggressive Mean Reversion) (Li, Zhao, Hoi, & Gopalkrishnan, 2012)
- CWMR (Confidence Weighted Mean Reversion) (Li, Hoi, Zhao, & Gopalkrishnan, 2011)
- Robust Median Reversion (RMR) (Huang, Zhou, Li, Hoi, & Zhou, 2016)

Reinforcement learning algorithm:

- PPO (Proximal Policy Optimization) (Schulman et al., 2017)
- SAC (Soft Actor Critic) (Haarnoja et al., 2018)
- A2C (Advantage Actor Critic) (Mnih et al., 2016)

Additionally, four versions of QFPIS models are implemented to determine the appropriate QPLs number of to be incorporated in risk-control mechanism.

- In QFPIS-1, the risk-control agent (PG agent) can perform below decisions:

1. Hold current order.
2. Close activating order at QPL + 1.

- In QFPIS-2, the risk-control agent (PG agent) can perform below decisions:

1. Hold current order.
2. Close activating order at QPL + 1.
3. Close activating order at QPL + 2.

For QFPIS-3 and QFPIS-4, an additional action is available for closing the activating order at QPL + 3 and QPL + 4, respectively.

5.2. Training analysis of QFPIS system

This section provided a training progress analysis of QFPIS system. To prevent overfitting, a validation set is used to calculate the cumulative return at each training episode for early stopping, and the best model is saved based on the cumulative return on the validation set as illustrated in Fig. 5. It showed that the cumulative return improved as the training progressed with converging around training episode 60.

Fig. 6 illustrated the weight changes of individual stocks in QFPIS-1 portfolio on Forex validation set on December 23, 2019. It showed that at the beginning of the training process, the model allocated a significant weight amount to EURUSD. As the model converged, the portfolio shifted its focus to trading AUDCAD and EURAUD, with less weight given to EURUSD.

Fig. 7 illustrated PG Agent actions of QFPIS-1/2/3/4 on Forex validation set. It is noted that the changes in PG Agent's actions. As the training progressed, the PG Agent converged, and likely to generate risk control actions on QPL + 1 and QPL + 2, rather than holding the current order or implementing risk control on QPL + 3 and QPL + 4. This may be due to QPL + 1 and QPL + 2 are more frequently in the range of daily high and low prices. It also showed that.

PG Agent's performance improves as the training progressed, along with the DDPG agent.

5.3. Performance evaluation

To evaluate and compare models' performances, several metrics are used, including Annual Return (AR), Cumulative Return Rate (CRR), Sharpe Ratio (SR), Maximum Drawdown (MDD), and final Accumulated Portfolio Value (fAPV). These performance metrics are listed in Tables 2-4 and Figs. 8-10 below.

As illustrated in Table 2 and Fig. 8, The results from Forex market showed that the QFPIS outperformed other baseline models as follows:

1. In fAPV column, QFPIS-4 achieved the best performance, the portfolio value at the end of the period is approximately twice the starting level, which indicated that QFPIS profitability is significant.
2. In Sharpe Ratio column, all four versions of QFPIS achieved high levels compared to other baseline models, which indicated QFPIS is a balanced selection to deal with the high risks in the forex market.

- Furthermore, the experimental data showed that the performance of QFPIS-4 > QFPIS-3 > QFPIS-2 > QFPIS-1 in terms of return, although the Sharpe ratio decreased, indicated that after incorporating more QPLs in risk control mechanism, the action space had contribution to PG Agent's exploration to expand the portfolio value.

The results for U.S. stock market as shown in Table 3 and Fig. 9 are:

- QFPIS-1/2/4 achieved high returns in testing period indicating its robustness, adaptability and profitability.
- The baseline models outperformed better in U.S. stock market than forex market, where all baseline models were unable to produce a positive return. In the U.S. stock market, A2C, OLMAR, PAMR, and RMR generated comparable performance to QFPIS-1 and QFPIS-2.

QFPIS performance in the China A-share market shown in Table 4 and Fig. 10 are:

- QFPIS-2 achieved the highest performance in China A-share market, indicating its adaptability. However, QFPIS-1/3/4 did not produce satisfactory performance due to liquidity and volatility in China market are lower than U.S. stock market. Additionally, training data length is the shortest in China A-share market, which means that QFPIS is prone to overfitting on the training data. In this case, more fine-tuning work is required to improve QFPIS performance in China A-share market.
- OLMAR/PAMR/CWMR/RMR performance was average in China A-share market, indicating that online portfolio selection strategies are a good option when historical data are limited.
- DDPG and A2C models generated positive returns in China A-share market, while SAC/PPO did not generate a positive return in any market indicating that DDPG and A2C are more suitable for portfolio management tasks.

5.4. Ablation study

This section conducted an ablation study on DDPG and A2C. It was noted in Section 5.3 that A2C could generated good performance in portfolio management, so it had included it in ablation study as listed in Table 5. It shows an interesting fact that combining A2C with QPL risk improved performance. For DDPG, as more QPL levels were added for risk control, the cumulative return increased, while the Sharpe ratio decreased. Fig. 11 showed the cumulative return rate and Sharpe ratio results from QFPIS-1 to QFPIS-8. It is evident that QFPIS-4 achieved the highest cumulative return rate. After QPL4, adding more QPLs did not contribute to performance significantly. Using QFPIS-4 in actual trading would be ideal as in (Qiu, Qiu, Yuan, Chen, & Lee, 2021).

6. Conclusion and future work

This paper presented a significant analysis of Deep Reinforcement Learning (DRL) in automatic trading and proposes a DRL system with a Quantum Finance Theory-inspired Profitable Portfolio Investment Management System (QFPIS). The experimental results showed that QFPIS outperformed other actor-critic-based RL methods and mainstream online portfolio selection strategies in Forex and the U.S. stock market.

However, QFPIS has limitations for improvement. These include:

Despite PG agent always trained for making trading decision to optimize the expected portfolio value, the portfolio value used in computation has the same operations for all products. (e.g., when QPL + 1 is selected, each product is attempted to close order at its own QPL + 1) It is necessary to separate operations with multi-agent mechanism integration enabling trading decision attributed to each product.

QFPIS only considered long positions. It is essential to achieve long and short positions simultaneously to meet real-life trading conditions.

QFPIS only followed the reward function setting from previous works (Jiang et al., 2017), where the reward function always provided positive/negative feedback in bullish/bearish phase, new reward function should be formulated.

The entire system is trained and tested in a virtual environment, the logic of placing orders and handling expectations is not considered. QFPIS can be developed to support trading strategies to real trading environment.

CRedit authorship contribution statement

Yitao Qiu: Conceptualization, Methodology, Visualization, Writing – original draft. **Rongkai Liu:** Writing – review & editing. **Raymond S. T. Lee:** Supervision, Project administration, Funding acquisition, Writing – review & editing.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

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